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Subject: Research methodology

Q1) Explain the meaning, objectives and characteristics of research.

Meaning of research:

Research refers to the systematic and objective search for knowledge. It involves careful investigation to discover new facts, solve problems and verify existing knowledge.

It includes defining problems, forming hypotheses, collecting and analyzing data and drawing conclusions.

Objectives of Research:

To gain new insights into a phenomenon (exploratory research).

To describe characteristics of individuals or situations (descriptive research).

To determine frequency, or relationship between variables (diagnostic research).

To test hypotheses and causal relationships (hypothesis-testing research).

Characteristics of research:

Controlled: minimizes the effect of external factors.

Systematic: Follows a logical sequence of steps.

Rigorous: Uses appropriate and well-justified procedures.

Empirical: Based on real world observations and evidence.

valid and verifiable: Results are accurate and can be confirmed.

critical: Methods and findings are carefully examined and evaluated.

Q3) Differentiate between research methods and research methodology. Also explain any two types of research:

Differentiate between research methods and research methodology:

Research Methods: Refers to the techniques and tools used for data collection (e.g., surveys, interviews, experiments). It is practical, specific and concrete.

Research Methodology: Refers to the systematic and scientific approach to solving research problems. It includes the theoretical framework, logic and overall strategy guiding the research.

Key differences:

- Methods focus on how data is collected, whereas methodology focuses on why and how research is conducted systematically.
- Methods are practical, methodology is theoretical.
- Methods are a part of methodology, while methodology covers the entire research process.

Types of Research:

Descriptive Research:

- Describe the current state of affairs.
- No control over variables; only observe and report facts.
- Uses surveys, comparative and correlational methods.

Applied Research:

- Aims to find solutions to real life problems.
- Focuses on practical application in society or business.
- Example: market research, social or economic problem solving.

Q9) what is a research problem? Explain the steps involved in defining a research problem.

Research problem:

A research problem refers to a difficulty or issue faced by a researcher in a theoretical or ^{practical} situation, which requires a systematic solution.

It arises when there are multiple alternatives available to achieve an objective and the researcher is uncertain about the best course of action.

Steps in defining a research problem:

Statement of the problem:

Define the problem in a broad and general term based on practical or scientific interests.

Understanding the nature of the problem:

Analyse the origin, objectives and environment of the problem through discussion and analysis.

Survey of literature:

Review existing studies, theories and data to gain knowledge, identify gaps, and refine the problem.

Developing ideas through discussions:

Discuss with experts and experienced persons.

to gain insights and improve clarity.

Rephrasing the problem:

convert the problem into a clear, specific and operational form to make it suitable for research and hypotheses development.

Q4) Discuss the importance of literature review in research and distinguish between primary and secondary sources.

Importance of literature review:

Literature review is a crucial step in research as it provides a clear understanding of existing knowledge on the topic. It helps the researcher to build a strong foundation before defining the research problem.

- Helps to gain background knowledge and understand key concepts.
- Identifies gaps, limitations and unanswered questions in previous studies.
- Avoid duplication of research and ensures originality.
- Assists in formulating research objectives, questions and hypotheses.
- guides the selection of research design, methods and techniques.
- Ensures the research is relevant, feasible and scientifically sound.

Primary and secondary sources:

primary sources:

- data collected first-hand by the researcher
- collected through methods like observation, interviews, questionnaires, surveys.

original and specific to the research problem.

Secondary sources:

- Data already collected and analysed by others.
- sources include books, journals, reports, government publications, records.
- must be checked for reliability, suitability and adequacy before use.

Q5) Explain the concept of research design and discuss its need and features of a good research design.

concept of research design:

Research design refers to the blueprint or plan of a research study. It is the arrangement of conditions for collection and analysis of data in a way that ensures relevance to the research objective with economy in time, effort and cost.

It defines what, where, when, how and by what means the research will be conducted. It provides a conceptual structure and includes decisions related to data collection, sampling, analysis and reporting.

need for Research design:

- Ensure smooth and systematic conduct of research.
- Helps in efficient use of time, effort and money.
- provides advance planning of methods and ^{tech} techniques.
- Increases reliability and accuracy of results.
- Helps to avoid errors and misleading conclusions.
- Assists in organizing ideas and identifying flaws before execution.

Features of a good research design:

- minimizes bias and maximizes reliability of data.
- ensures maximum information with minimum error.
- should be flexible, appropriate and efficient.
- clearly defines research problem and objectives.
- specific data collection methods and techniques.
- considers time and cost constraints.
- Based on factors like availability of resources, skills and nature of the problem.

Q6) what is sampling design? Explain the steps involved and types of sample design.

Sampling design:

sampling design is a definite plan for obtaining a sample from given population. It refers to the procedure or technique used to select items for the sample and also determines the size of the sample before data collection.

Steps in sampling design:

Type of universe:

clearly define the population (finite or infinite) to be studied.

sampling unit:

decide the unit of selection such as individuals, family, or area.

source list:

prepare a complete and reliable list of population elements.

size of sample:

determine an optimum sample size considering accuracy, population size and cost.

parameters of interest:

Identify the population characteristics to be studied.

Budgetary Constraints:

Consider cost and available resource.

sampling procedure:

select the appropriate sampling technique.

Types of sample design:

Non probability sampling:

- selection is based on researcher's judgment
- No equal chance for all elements.
- may involve bias and cannot estimate sampling error.

examples: purposive sampling, quota sampling.

probability sampling:

- Based on random selection.
- Each elements has an equal chance of inclusion.
- produces reliable and representative results.
- Allows measurements of sampling error and statistical analysis.

Q7) Explain the stages in the scientific research process.

The scientific research process involves a series of systematic steps to solve a problem and generate knowledge.

stages in scientific research process:

Identification of research problem:

clearly define the problem or issue to be studied.

Review of literature:

study existing research to gain background knowledge and identify gaps.

Formulation of hypothesis:

develop a testable assumption or prediction about the problem.

Research design:

prepare a plan or blueprint for data collection and analysis.

Data Collection:

gather relevant data using methods like observation, interviews, surveys.

Data analysis:

process and analyse data using statistical or logical techniques.

Interpretation of results:

draw meaningful conclusions from the analysed data.

Testing of hypothesis:

verify whether the hypothesis is accepted or rejected.

Report writing:

present findings in a structured and systematic report.

Q8) Discuss the process of hypothesis testing and interpretation of data in research methodology.

Hypothesis testing process:

Hypothesis testing is a scientific method of verifying assumptions about relationships between variables.

Formulation of Hypothesis:

Develop null hypothesis (H_0) and alternative hypothesis (H_1).

Selection of Significance Level:

Choose a level of significance (eg 0.05) to decide acceptance/rejection.

Selection of Test Statistic:

Select appropriate statistical test (t-test, chi-square, etc).

Data collection:

Gather relevant and reliable data.

Calculation of Test Statistic:

Compute the value using collected data.

Decision making:

Compare with critical value and accept or reject H_0 .

Interpretation of Data:

Interpretation involves drawing meaningful conclusions from analysed data.

Convert analysed data into useful information.

Helps in understanding relationships and patterns.

- provides basis for conclusions, and decision-making.
- ensures findings are linked with objectives and hypothesis.
- Must be logical, unbiased and clear.

Thus hypothesis testing and interpretation together ensure that research findings are valid, reliable and meaningful.